

The University of the South Pacific

School of Computing, Information & Mathematical Sciences

CS412 Artificial Intelligence

Lab 2 (worth 4%)

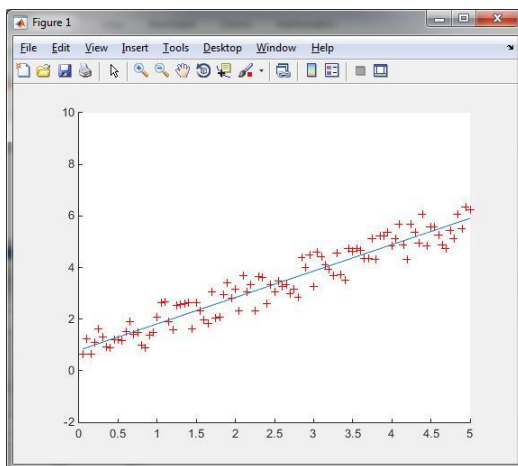
This assessment covers the following course learning outcomes partially:

CLO 1 – Evaluate machine learning algorithms and their limitations

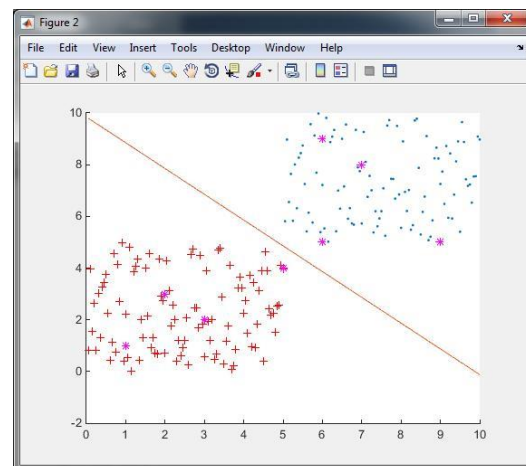
CLO 3 – Apply machine learning algorithms to solve real life complex computing problems.

In this lab you will understand how linear regression works that you have learnt in your lecture week 3. Do the following questions:

- 1 - Read and understand linear regression.
- 2 - Randomly generate data in 2D plane and determine the best fit line using linear regression formula [$W = \text{pseudoinverse}(X) * Y$].
- 3 - Draw the best fit line together with data points. What is your sum of square Error (E_{in})?
- 4 - Test your program.
- 5 - Now generate random data as two separate clusters in 2D plane.
- 6 - Scatter plot your data. See the sample graphs below



Regression



Classification

7 - Using linear regression for classification classify your data into two classes using the same formula given above. [Note: this time you have to label your data with +1 and -1 to indicate two different classes]. Does it give you an expected solution? Try different class sizes.

8 - Draw the separation plane if possible. [You need to determine the linear equation $W'X=0$ or $Xw = 0$]

9 - Test your program.

10 – Show how linear regression for classification can improve pocket algorithm with PLA.

BOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Abstraction & Design	Unsatisfactory II. Irrelevant prototyping VIII. Poor architecture and design makes future updates hard to incorporate.	Satisfactory I. Analysis of requirements such as risk analysis & mitigation, prototyping, identification of true/false requirements etc. III. all relevant constraints and requirements are mentioned VI. Appropriate choice of the architecture and design	Good I. All satisfactory and ... II. Choice of architecture and design is based on robustness, scalability, evolution and long term management. III. Innovative or creative design of algorithms (efficiency, structure and data structures)	1	
Programming	Unsatisfactory I. Plagiarism II. Poor indentation, hard to read and follow the code III. Lots of bugs and/or errors IV. Program produces unexpected output VII. Hard coding of data in the program. VIII. Poor structure	Satisfactory I. Able to write a complex code for an open ended problem II. Use of highest standard of programming practices such as modularity with appropriate architecture, low	Good I. All satisfactory and... demonstrate very good programming skills.	1	

		coupling, high cohesion, standard docs (eg. Javadoc) etc. III. Appropriate testing is done.			
Data & Information Management	Unsatisfactory VII. Very limited or undesirable data has been collected IX. Poor or incorrect inference is derived from the collected/given data	Satisfactory VII. Appropriate data has been collected IX. logical inference is derived from the collected/given data	Good I. All satisfactory and III. Extensive analysis is done on the collected/given data. IV. Exceptional Inference is derived from the data.	2	
Sub Total & comments				4	